

Standard Scaler  $\rightarrow$  feature scaler, standardize  
by removing the mean  
and scaling to unit variance.

Class weight  $\rightarrow$  imbalanced  
dataset go to

MLP Classifier  $\rightarrow$  Multi Layer Perceptron

Ordinal Encoder  $\rightarrow$  Categorical value  $\rightarrow$  Numerical value  
single colour with integers

One hot Encoding  $\rightarrow$  multiple col with 0/1

ROC Curve  $\rightarrow$  True Positive vs False positive

(TP)  $\rightarrow$  Correctly predicts a positive outcome

$$\frac{TP}{TP + FN}$$

$$\frac{FP}{FP + TN}$$

(FN)  $\rightarrow$  incorrectly predicts a negative outcome

(AUC)  $\rightarrow$  Area under curve  $\rightarrow$  ROC metric

Confusion Matrix  $\rightarrow$  True vs predicted class

Accuracy Score  $\rightarrow$   $\frac{\text{correct prediction}}{\text{Total pred.}}$

$$\text{Precision} = \frac{\text{Positive}}{\text{Actually Positive}}$$

$$F1 \text{ score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\frac{TP}{TP + FN}$$

Sequential → stack layers  
to build NN

Perce → "fully connected  
neural network"

## Data Splitting

[ a, b, c, d ]

col row  
cut numerical  
row star list a from

categorical 3 same

·T → transpose

·var() → variance → data right spread out  
→ 0 (Right)

-skew → skewed

20 (Left)  
→ deciding  
star left.

·unique → no. of unique values in a col.

## Correlation Analysis

Confusion matrix

• corr

Age vs Job level = 0.179

Age vs Distance = -0.049

NAN → no correlation

① →

perfect correlation

⊖

## Correlations using diff methods

Pearson → linear

Spearman → rank based.

SMOTE → imbalance dataset → balance

## Models

Sequential → feed forward NN

relu →  $\max(0, x)$

Dim (X), a  
↓  
neurons

input\_dim = X.shape[1]

[1]

epoch 3 / 100 → Total  
59 no. iterations

• shape [0] → no. of samples

• shape [1] → no. of features

$$G = \frac{1}{1 + e^{-z}}$$

↓  
Binary classifier

train scaled, y - sampled

epoch = 100  
 100 Times Run

scale  
 Addition  
 re-sample

batch size = 64

batch = 64

64 for sample  
 process weight update

⇒ Mo y-pred-nn → Test go for run  
 > 0.5  
 raw values  
 0.67  
 0.73  
 0.21

[1] shape X = 24000 [1]  
 input dim = 24000 [1]

$$e = \frac{1}{1 + e^{-x}}$$

sigmoid

shape [0] → no. of samples  
 shape [1] → no. of features